CropState: Image-Driven Confidence-Aware Growth-Stage-Gated Retrieval for Rice Decision Support

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***Abstract*—Agricultural management actions depend strongly on crop development, yet many decision-support systems retrieve evidence using only lexical or semantic similarity. A passage may therefore be topically relevant while being inappropriate for the current rice growth stage. This paper proposes CropState, an image-driven, confidence-aware growth-stage-gated retrieval framework for rice decision support. A vision model first processes a field image and produces a probability distribution over six BBCH-derived rice macro-stages. Rather than converting this distribution immediately into a single hard label, CropStatepreserves uncertainty, optionally fuses it with a temporal prior, and adaptively controls the contribution of stage compatibility during retrieval. The downstream component uses fixed care topics rather than free-form chatbot questions, combines BM25 and dense retrieval using normalized Reciprocal Rank Fusion, and returns structured stage-specific evidence. The study addresses two research questions: how accurately rice growth stages can be recognized from field images, and whether confidence-aware soft gating is more robust than top-1 hard filtering when image-based stage predictions are uncertain or incorrect. The protocol includes image-classification baselines, calibration analysis, oracle-stage retrieval, end-to-end image-based retrieval, the Stage-Inappropriate Retrieval Rate, robustness degradation, ablation studies, and paired statistical analysis. Empirical claims are intentionally deferred until the image dataset and held-out experiments are completed.**

***Index Terms*—rice growth stage, computer vision, BBCH, image classification, retrieval-augmented generation, uncertainty-aware retrieval, agricultural decision support**

I. Introduction

Agricultural recommendations are conditional on crop development. Water management, nutrient application, disease risk, weed control, and harvest preparation can change substantially between establishment, tillering, reproductive development, grain filling, and ripening. A retrieval system that ignores growth stage may return evidence that is semantically related to the topic but mistimed in agronomic application.

A practical system must also determine the current stage without requiring a user to formulate a technical question. Field images provide a natural input because phenological development is partially observable from plant morphology. However, image-based stage recognition is uncertain, especially near stage boundaries, under poor lighting, or when images from different fields and seasons vary in background and scale. Converting an uncertain visual prediction into a single stage label and applying a hard metadata filter may amplify classification errors by removing all evidence associated with the true stage.

This paper proposes CropState, an image-driven pipeline that connects rice growth-stage recognition to stage-aware retrieval. The vision model outputs a distribution rather than only a top-1 class. This distribution becomes a stage belief that controls how strongly the retriever favors stage-compatible evidence. When the visual prediction is concentrated and calibrated, stage compatibility receives more weight. When it is diffuse or conflicts with a temporal prior, the system relaxes stage conditioning to preserve recall.

The system is not designed as a free-form chatbot. After an image is uploaded, it automatically retrieves evidence for a fixed set of care topics such as water, nutrients, disease risk, pest risk, and weed management. Outputs are structured as detected stage, confidence, stage-specific recommendations, and supporting sources.

The intended contributions are:

1) an image-driven pipeline that links six-class rice growth-stage recognition to structured agricultural retrieval;
2) a confidence-aware soft-gating method that uses the full visual probability distribution instead of only the top-1 stage;
3) an end-to-end evaluation protocol that measures how vision uncertainty and classification errors propagate to retrieval quality and stage-inappropriate evidence.

II. Related Work

A. Retrieval-Augmented Generation and Hybrid Retrieval

Retrieval-Augmented Generation grounds model outputs in external evidence [1]. Sparse retrieval such as BM25 captures lexical correspondence [2], while dense retrieval captures semantic similarity [3]. Reciprocal Rank Fusion (RRF) combines ranked lists without directly comparing heterogeneous raw scores [4]. CropStateuses BM25 and dense retrieval for candidate generation, then applies confidence-aware stage re-ranking.

B. Agricultural Context-Aware Retrieval

Crop GraphRAG organizes crop-protection knowledge around crop–pest–disease–symptom–control relations and per-

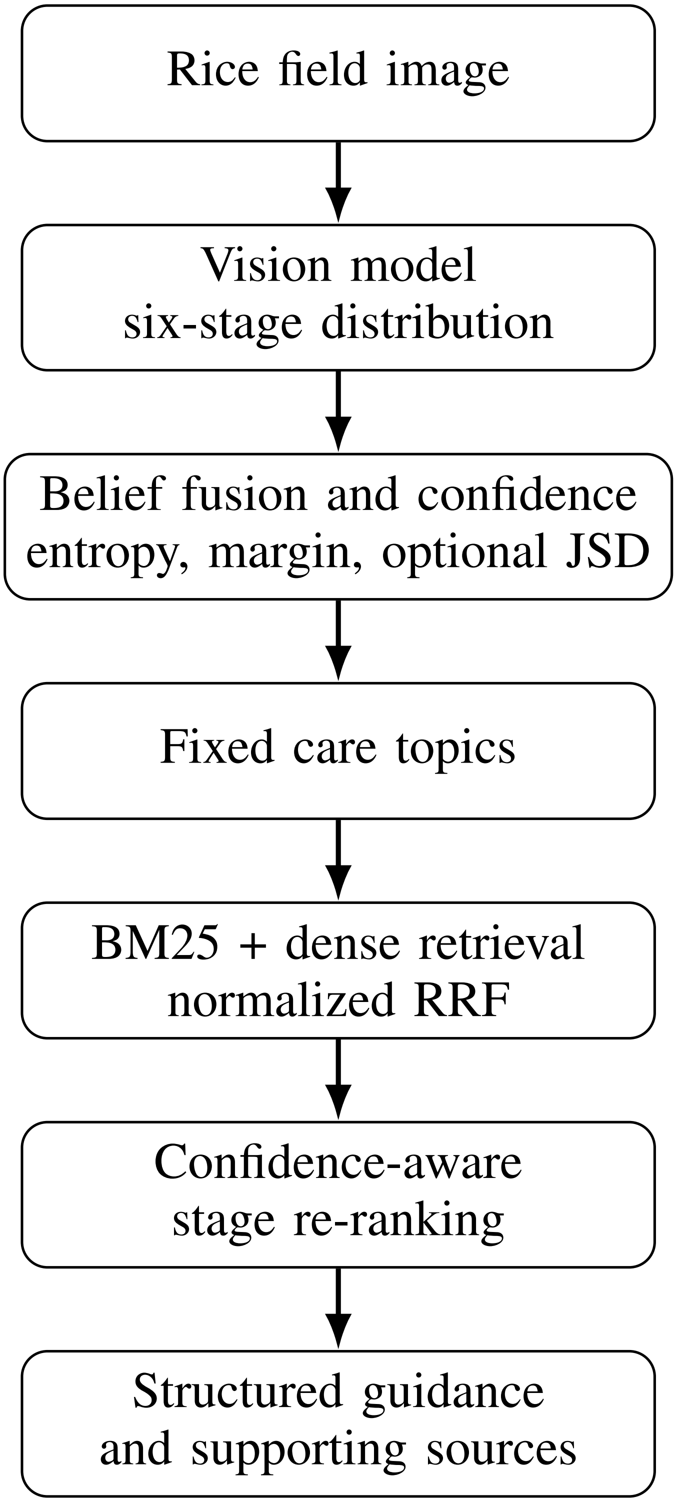

Fig. 1. Image-driven CROPSTATEpipeline without free-form chatbot interaction.

forms graph-based evidence aggregation. Its primary contribution is relational knowledge structure rather than uncertainty in a visually estimated crop state. AgriRegion conditions retrieval on geographic metadata to reduce recommendations that are valid in one region but inappropriate in another. In contrast, CROPSTATEconditions retrieval on an ordered phenological state inferred from images and explicitly evaluates robustness when that state estimate is wrong. Table I summarizes the distinction.

C. Rice Growth-Stage Recognition

The BBCH scale provides standardized phenological codes for crop development [5], [6]. Prior work has shown that machine-learning methods can recognize rice growth stages from imagery [7]. Modern image classifiers commonly use pretrained convolutional or transformer backbones, and calibration methods such as temperature scaling can improve the relationship between predicted confidence and empirical correctness [8]–[10].

The present work does not claim a new vision architecture. The scientific focus is the interface between visual stage uncertainty and retrieval. The vision model is an upstream estimator whose full output distribution is retained for downstream decision support.

III. PROBLEM FORMULATION

A. Growth-Stage Space

Let $\Phi = \langle \phi_1, \dots, \phi_m \rangle$ denote an ordered set of rice growth stages. The initial study uses the six macro-stages in Table II. These groups are an experimental abstraction of BBCH codes rather than a new phenological standard. The primary task is whole-image classification. Object detection is outside the main evaluation and may only be used as an auxiliary localization method.

B. Image-Based Stage Belief

For rice image I_t, a vision model with parameters θ produces logits $\mathbf{o}_t = f_\theta(I_t)$. The visual stage distribution is

$$p_{\mathrm{vis},t} = \mathrm{softmax}(\mathbf{o}_t / T), \tag{1}$$

where $T > 0$ is a temperature parameter estimated on the validation set. The top-1 prediction is

$$\widehat{\phi}_t = \arg\max_i p_{\mathrm{vis},t}(\phi_i), \tag{2}$$

but CROPSTATEretains the complete distribution instead of discarding non-maximal stages.

C. Optional Temporal Prior and Fusion

If sowing or transplanting time is available, a temporal vector $p_{\mathrm{time},t}$ can be constructed from agronomic development intervals. A transition matrix $A \in [0,1]^{m \times m}$ defines plausible progression from a previous belief B_{t-1}:

$$\widetilde{B}_t = A^\top B_{t-1}. \tag{3}$$

Given prior weight w_p and evidence weights w_e, log-linear fusion is

$$z_i = w_p \log(\widetilde{B}_t(\phi_i) + \epsilon) + \sum_{e \in \mathcal{E}_t} w_e \log(p_e(\phi_i) + \epsilon), \tag{4}$$

followed by

$$B_t(\phi_i) = \frac{\exp(z_i)}{\sum_j \exp(z_j)}. \tag{5}$$

TABLE I
POSITIONING RELATIVE TO CLOSELY RELATED RETRIEVAL DIRECTIONS.

Direction	Primary context	Main retrieval mechanism	Difference from CROPSTATE
Standard RAG	Query content	Sparse/dense passage retrieval	Does not model phenological applicability.
Crop GraphRAG	Crop–pest–symptom–control relations	Local/global graph retrieval	Focuses on relational structure, not image-derived stage belief or stage-error robustness.
AgriRegion	Geographic region and local sources	Region-prioritized re-ranking	Uses geographic context rather than ordered visual phenology.
CROPSTATE	Image-derived rice stage belief	Confidence-adaptive soft re-ranking	Evaluates propagation of visual uncertainty into stage-specific retrieval.

TABLE II
RICE MACRO-STAGES USED IN THE PLANNED BENCHMARK.

Group	BBCH	Label
Germination and leaf development	00–19	Establishment
Tillering	20–29	Tillering
Stem elongation and booting	30–49	Stem/booting
Heading and flowering	50–69	Reproductive
Grain development	70–79	Grain filling
Ripening	80–89	Ripening

In the image-only configuration, $B_t = p_{\mathrm{vis},t}$. In the fused configuration, $\mathcal{E}_t$ contains visual and temporal evidence.

D. Confidence and Source Agreement

Belief concentration is measured using normalized entropy:

$$\Gamma_t = 1 - \frac{H(B_t)}{\log m}, \qquad H(B_t) = -\sum_i B_t(\phi_i) \log B_t(\phi_i). \tag{6}$$

A complementary top-two margin is

$$M_t = B_t^{(1)} - B_t^{(2)}, \tag{7}$$

where $B_t^{(1)}$ and $B_t^{(2)}$ are the largest and second-largest belief values.

When both visual and temporal distributions are available, agreement is

$$G_t = 1 - \frac{\mathrm{JSD}(p_{\mathrm{vis},t}, p_{\mathrm{time},t})}{\log 2}. \tag{8}$$

The confidence proxy is defined as

$$\kappa_t = \eta_1 \Gamma_t + \eta_2 M_t + \eta_3 G_t, \tag{9}$$

where $\eta_1 + \eta_2 + \eta_3 = 1$. In the image-only configuration, $\eta_3 = 0$. This confidence is a routing signal, not proof that the predicted stage is correct.

E. Documents and Stage Compatibility

Let $\mathcal{D}$ be a collection of agricultural knowledge chunks. Each chunk d includes source provenance, care topic, applicability metadata, and a compatibility vector

$$C(d, \phi_i) \in [0, 1]. \tag{10}$$

A value of one denotes direct applicability, an intermediate value denotes adjacent-stage or multi-stage applicability, and zero denotes incompatibility or explicit contraindication. Generic and unknown applicability are represented separately.

F. Research Questions

G. Image Dataset and Split Protocol

Each image record stores an image identifier, parent-image identifier, field, capture session, date, season, variety, days after sowing, BBCH label, macro-stage, source, license, and annotation status. Image patches cropped from the same parent image are not treated as independent samples. All overlapping patches, burst images, and images from the same field-session group are assigned to a single split. The principal evaluation uses a field- or season-held-out test set to reduce background and acquisition leakage. Random image-level splitting is reported, if at all, only as a diagnostic comparison.

Images are accepted as ground truth only when stage-relevant morphology is visible. Top-down patches may support establishment and tillering recognition, but later stages require sufficient evidence of stem elongation, booting, panicle emergence, grain development, or ripening. Images that omit decisive structures are marked uncertain or excluded from supervised evaluation. At least a substantial subset is independently annotated by two reviewers and disagreements are adjudicated before the test set is locked.

IV. PROPOSED IMAGE-DRIVEN METHOD

A. Automated Retrieval Topics

The system does not require free-form user questions. After stage inference, it retrieves evidence for a predefined set of care topics:

$$\mathcal{T} = \{\text{water}, \text{nutrients}, \text{disease}, \text{pests}, \text{weeds}\}. \tag{11}$$

Each topic $\tau \in \mathcal{T}$ has a fixed query template such as "rice water-management guidance" or "rice disease risk and preventive management." Stage information is applied through the re-ranking mechanism rather than being treated as a conversational answer rule.

B. Hybrid Candidate Generation

For topic query q_τ, BM25 and dense retrieval produce ranked lists L_b and L_d. Their RRF score is

$$S_{\mathrm{rrf}}(x, q_\tau) = \frac{1}{k_0 + r_b(x)} + \frac{1}{k_0 + r_d(x)}. \tag{12}$$

Because RRF and stage scores have different ranges, RRF is min–max normalized over candidate union $U = L_b \cup L_d$:

$$\overline{S}_{\mathrm{base}}(x) = \frac{S_{\mathrm{rrf}}(x) - \min_{u \in U} S_{\mathrm{rrf}}(u)}{\max_{u \in U} S_{\mathrm{rrf}}(u) - \min_{u \in U} S_{\mathrm{rrf}}(u) + \epsilon}. \tag{13}$$

TABLE III
RESEARCH QUESTIONS AND HYPOTHESES.

ID	Research question	Hypothesis
RQ1	How accurately can a vision-based model recognize six rice macro-stages from field images collected across held-out fields or seasons?	H1: A fine-tuned pretrained vision model achieves higher macro-F1 and lower mean absolute stage distance than majority and shallow-feature baselines.
RQ2	Does confidence-aware soft gating produce more robust stage-specific retrieval than top-1 hard filtering when image-based stage predictions are uncertain or incorrect?	H2: Adaptive soft gating exhibits lower nDCG degradation and a smaller SIRR increase than hard filtering on misclassified and low-confidence test images.

C. Belief-Weighted Stage Score

The expected compatibility of document x is

$$S_{\text{stage}}(x, B_t) = \sum_{i=1}^{m} B_t(\phi_i) C(x, \phi_i). \tag{14}$$

Let $\overline{S}_{\text{source}}(x) \in [0, 1]$ be a normalized provenance or review-status score. Stage weight is adapted by

$$\beta_t = \beta_{\min} + (\beta_{\max} - \beta_{\min})\kappa_t, \tag{15}$$

and $\alpha_t = 1 - \beta_t - \mu$. The final score is

$$S(x, q_\tau, B_t) = \alpha_t \overline{S}_{\text{base}}(x) + \beta_t S_{\text{stage}}(x, B_t) + \mu \overline{S}_{\text{source}}(x). \tag{16}$$

The proposed method does not exclude a document solely because it is incompatible with the top-1 stage. Exclusionary filtering is retained as a comparison baseline.

V. RESEARCH METHODOLOGY

A. Image Dataset

Each image record should include image identifier, field identifier, season, variety when available, capture date, days after sowing or transplanting, BBCH code, macro-stage, camera distance, and annotation status. Stage labels should be based on observable morphology and reviewed by an agronomic expert or trained annotator.

To reduce leakage, train, validation, and test splits must be grouped by field, farm, or season. Near-duplicate images from the same plot must not be distributed across splits. The final study should report image counts per stage, number of fields, class imbalance, acquisition conditions, and the proportion receiving independent review.

B. Knowledge Base and Retrieval Labels

The document collection should contain traceable agronomic sources. Each chunk includes care topic, stage range, applicability type, source provenance, authority score, and contraindications. Evaluation labels identify relevant chunks and stage-inappropriate chunks for each fixed care topic and ground-truth image stage.

C. Vision Baselines

The vision study compares at least:

- majority-class prediction;
- a shallow hand-crafted-feature or small-CNN baseline;
- pretrained ResNet-18;
- pretrained EfficientNet-B0 or another lightweight modern backbone.

All learned models use the same field-grouped splits and evaluation protocol. Hyperparameters are selected on the validation set. Temperature scaling is fitted after model training using validation logits only.

Algorithm 1 Image-driven confidence-aware retrieval

Require: Rice image I_t, optional prior B_{t-1}, optional time evidence $p_{\text{time},t}$, corpus $\mathcal{D}$, topic set $\mathcal{T}$
Ensure: Stage belief B_t, confidence κ_t, ranked evidence for each topic

$p_{\text{vis},t} \leftarrow \text{softmax}(f_\theta(I_t)/T)$
if time evidence is available **then**
  $\widetilde{B}_t \leftarrow A^\top B_{t-1}$
  $B_t \leftarrow \text{LOGLINEARFUSE}(\widetilde{B}_t, p_{\text{vis},t}, p_{\text{time},t})$
  $G_t \leftarrow 1 - \text{JSD}(p_{\text{vis},t}, p_{\text{time},t})/\log 2$
else
  $B_t \leftarrow p_{\text{vis},t}$; $G_t \leftarrow 0$
end if
$\Gamma_t \leftarrow 1 - H(B_t)/\log m$
$M_t \leftarrow B_t^{(1)} - B_t^{(2)}$
$\kappa_t \leftarrow \eta_1\Gamma_t + \eta_2 M_t + \eta_3 G_t$
for all $\tau \in \mathcal{T}$ **do**
  $L_d \leftarrow \text{DENSERETRIEVE}(q_\tau, \mathcal{D})$
  $L_b \leftarrow \text{BM25RETRIEVE}(q_\tau, \mathcal{D})$
  $U \leftarrow L_d \cup L_b$; normalize RRF over U
  for all $x \in U$ **do**
    $S_{\text{stage}} \leftarrow \sum_i B_t(\phi_i)C(x, \phi_i)$
    $\beta_t \leftarrow \beta_{\min} + (\beta_{\max} - \beta_{\min})\kappa_t$
    $S(x) \leftarrow (1 - \beta_t - \mu)\overline{S}_{\text{base}} + \beta_t S_{\text{stage}} + \mu\overline{S}_{\text{source}}$
  end for
  $L_\tau \leftarrow \text{SORTDESCENDING}(U, S)$
end for
return $B_t, \kappa_t, \{L_\tau\}_{\tau\in\mathcal{T}}$

D. Retrieval Baselines

All methods use the same corpus, chunking, embedding model, candidate depth, fixed topics, and top-k. Only the stage-conditioning mechanism differs.

E. Experiment 0: Vision Feasibility

Experiment 0 determines whether the available images support reliable six-stage classification. Models are compared

TABLE IV
RETRIEVAL METHODS USED FOR COMPARISON.

ID	Description
B0	Ungated BM25+dense retrieval fused with normalized RRF.
B1	Top-1 stage query expansion using the predicted image label.
B2	Hard filtering by the top-1 predicted image stage.
B3	Soft stage re-ranking with a fixed stage weight.
P	Proposed adaptive re-ranking using the complete image-derived belief and confidence.

using a field- or season-held-out split. The experiment reports accuracy, macro-F1, per-class recall, confusion matrix, Mean Absolute Stage Distance (MASD), Brier score, and Expected Calibration Error (ECE). If performance is close to the majority baseline or errors are dominated by leakage-sensitive background cues, the image protocol or stage grouping must be revised before end-to-end evaluation.

F. Experiment 1: Oracle-Stage Retrieval

Experiment 1 supplies the ground-truth stage from each image to all stage-aware retrieval methods. This isolates the value of stage conditioning from vision errors. It evaluates whether stage-aware methods reduce stage-inappropriate evidence under ideal stage information. B0, oracle B1, oracle B2, B3, and P are compared using the same retrieval candidates.

G. Experiment 2: End-to-End Image-Based Retrieval

Experiment 2 replaces oracle labels with the calibrated probability distribution produced by the selected vision model. Results are stratified by:

1) correctly classified high-confidence images;
2) correctly classified low-confidence images;
3) adjacent-stage misclassifications;
4) non-adjacent misclassifications;
5) optional visual–temporal conflicts.

The main comparison is between B2, B3, and P. B0 serves as a safety reference when visual evidence is unreliable.

H. Metrics

Vision metrics include accuracy, macro-F1, per-class recall, ECE, Brier score, and

$$\mathrm{MASD} = \frac{1}{N}\sum_{j=1}^{N} |\,\mathrm{index}(\widehat{\phi}_j) - \mathrm{index}(\phi_j^*)|. \tag{17}$$

Retrieval metrics include Precision@k, Recall@k, and nDCG@k. Stage-Inappropriate Retrieval Rate is

$$\mathrm{SIRR}\,@k = \frac{1}{k}\sum_{x\in\mathrm{TopK}} \mathbb{I}[C(x,\phi^*)=0]. \tag{18}$$

For a metric M that should be maximized, degradation from oracle to image-based retrieval is

$$\Delta M = M_{\mathrm{oracle}} - M_{\mathrm{image}}. \tag{19}$$

For SIRR, degradation is $\Delta\,\mathrm{SIRR} = \mathrm{SIRR}_{\mathrm{image}} - \mathrm{SIRR}_{\mathrm{oracle}}$.

I. Ablation and Sensitivity Analysis

Ablation removes one component at a time: temperature calibration, entropy term, margin term, visual–temporal agreement, adaptive weighting, adjacent-stage compatibility, source score, and RRF normalization. Sensitivity analysis varies $\beta_{\min}$, $\beta_{\max}$, μ, candidate depth, top-k, and compatibility values. These settings are tuned only on the validation or development set.

J. Statistical Analysis

All retrieval methods are evaluated on the same images and topics; therefore, paired analysis is required. The final study reports bootstrap 95% confidence intervals, paired permutation or Wilcoxon signed-rank tests, effect sizes, and Holm correction for multiple comparisons. Vision-model comparisons use paired predictions on the same held-out images. Results should also be reported by class and by confidence bin rather than only as aggregate means.

VI. RESULTS REPORTING PLAN

This section will be populated only after experiments are completed. Tables remain empty to pre-specify the reporting structure and prevent fabricated or selectively reported findings.

VII. DISCUSSION

A useful result does not require the proposed method to outperform every baseline under every condition. Hard filtering may achieve the best oracle-stage SIRR while failing sharply on misclassified images. Adaptive soft gating may sacrifice a small amount of oracle precision to preserve relevant evidence under visual uncertainty. Conversely, a calibrated image model can still be confidently wrong under field shift, in which case confidence-aware gating may continue to overweight an incorrect stage. Such cases should be reported explicitly.

If image recognition performs poorly, the study cannot attribute downstream failures solely to retrieval. Error analysis must distinguish morphological ambiguity, class imbalance, background leakage, camera-scale variation, metadata error, missing knowledge, and retriever failure. If oracle-stage retrieval does not reduce SIRR, the knowledge collection or stage annotation may be insufficiently stage-sensitive even if the vision model performs well.

VIII. THREATS TO VALIDITY

A. Construct Validity

An image may not reveal all phenological cues, and a single image may not represent an entire field. Stage compatibility also does not guarantee complete agronomic safety because region, variety, weather, regulation, and field condition remain relevant.

B. Internal Validity

Random image splitting can leak field background and acquisition conditions. The study therefore requires group-based splits. Differences in augmentation, class weighting, embedding models, chunking, or candidate depth may also confound comparisons and must be controlled.

TABLE V
EXPERIMENT 0: VISION RESULTS ON THE HELD-OUT IMAGE TEST SET (NOT YET POPULATED).

Model	Accuracy	Macro-F1	MASD ↓	ECE ↓	Brier ↓	95% CI
Majority class	–	–	–	–	–	–
Small CNN	–	–	–	–	–	–
ResNet-18	–	–	–	–	–	–
EfficientNet-B0	–	–	–	–	–	–

TABLE VI
EXPERIMENT 1: ORACLE-STAGE RETRIEVAL RESULTS (NOT YET POPULATED).

Method	P@5	R@5	nDCG@5	SIRR@5 ↓	95% CI of primary difference
B0: Ungated hybrid	–	–	–	–	–
B1: Oracle query expansion	–	–	–	–	–
B2: Oracle hard filter	–	–	–	–	–
B3: Fixed soft re-ranking	–	–	–	–	–
P: Adaptive re-ranking	–	–	–	–	–

TABLE VII
EXPERIMENT 2: END-TO-END IMAGE-BASED RETRIEVAL (NOT YET POPULATED).

Image condition	B0 Ungated	B2 Hard filter	B3 Fixed soft	P Adaptive
Correct, high confidence	–	–	–	–
Correct, low confidence	–	–	–	–
Adjacent-stage error	–	–	–	–
Non-adjacent error	–	–	–	–
Visual–temporal conflict	–	–	–	–

C. External Validity

A rice-only study cannot establish effectiveness for other crops. Results are limited to the varieties, fields, seasons, image viewpoints, and knowledge sources represented in the dataset.

D. Label Validity

Days after sowing are not absolute stage labels. Ground truth should be based on morphological criteria and expert review. Boundary cases may legitimately support adjacent-stage labels and should not be forced into a falsely precise class without documentation.

IX. ETHICAL AND SAFETY CONSIDERATIONS

The system is intended for decision support, not autonomous agronomic authority. Outputs involving pesticides, dosage, withholding periods, or regulated actions must be grounded in current authoritative sources. Low-confidence images should trigger uncertainty notices or a request for a clearer image rather than an overconfident recommendation. Dataset releases should remove precise field locations and personal identifiers. Images collected from farms require appropriate consent and data-governance procedures.

X. CONCLUSION

This paper proposes an image-driven version of CROP-STATEthat recognizes rice growth stage from field images and uses the complete visual stage distribution to control evidence retrieval. The method replaces brittle top-1 filtering with confidence-aware soft gating, uses fixed care topics rather than free-form chatbot interaction, and evaluates both vision quality and downstream retrieval robustness. The scientific claims will be determined through a held-out image-classification experiment, an oracle-stage retrieval experiment, and an end-to-end image-based retrieval experiment. No effectiveness claim is made before those evaluations are completed.

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